A projection-support formula for Hilbert–Schmidt closure

Candidate partial answer to Question 4.14 of arXiv:2511.23121

11 August 2026

Status: candidate partial result, likely valid. The Hilbert–Schmidt closure projection is exactly the complement of a join of support projections of trace-class test vectors. For $M = \mathcal{B}(H)$ this completely classifies all finite-corank projections, including a sharp corank-one dichotomy. The formula does not yet eliminate the trace-class test-vector condition for an arbitrary von Neumann algebra, so it is not claimed as a complete intrinsic classification in the full generality requested by the source.

1 Source question and notation

Let $M \subseteq \mathcal{B}(H)$ be a von Neumann algebra and put

$$K = \mathcal{S}_2(H), \qquad A = M \bar{\otimes} M^{\mathrm{op}} \subseteq \mathcal{B}(K).$$

Under the correspondence in Daws's paper, an HS quantum relation is $V = eK$ for a projection $e \in A$. Its Hilbert–Schmidt closure is

$$V^{\mathrm{hsc}} = (\overline{V}^{w^*}) \cap \mathcal{S}_2(H),$$

and corresponds to a projection $e^{\mathrm{hsc}} \in A$ with $e \le e^{\mathrm{hsc}}$. Question 4.14 asks for a description of $e \mapsto e^{\mathrm{hsc}}$ at the level of $M \bar{\otimes} M^{\mathrm{op}}$ [1, Question 4.14].

Write $D = \mathcal{S}_1(H) \subset K$. For $\xi \in K$, let

$$\omega_\xi(a) = \langle a\xi, \xi \rangle \quad (a \in A), \qquad s_A(\omega_\xi) \in A$$

denote the support projection of this normal positive functional.

2 The support-hull formula

Theorem 1 (Projection-support formula)**.** *For every projection $e \in A$,*

$$e^{\mathrm{hsc}} = 1 - \bigvee \{ s_A(\omega_\xi) : \xi \in \mathcal{S}_1(H),\ e\xi = 0 \}. \tag{1}$$

Equivalently, if

$$W_e = \overline{\mathcal{S}_1(H) \cap (1-e)K}^{\|\cdot\|_2},$$

then the orthogonal projection r_e onto W_e belongs to A and

$$e^{\mathrm{hsc}} = 1 - r_e. \tag{2}$$

In particular,

$$e = e^{\mathrm{hsc}} quad \Longleftrightarrow \quad \overline{\mathcal{S}_1(H) \cap \ker e}^{\|\cdot\|_2} = \ker e. \tag{3}$$

Proof intuition. The source identifies V^{hsc} as the Hilbert-space annihilator of the trace-class vectors normal to V. Thus its orthogonal complement is the closed trace-class part of $\ker e$. That subspace is automatically reducing for the commutant of A, so its projection lies in A. Support projections package the same closed subspace as a join inside the projection lattice of A.

Proof. Proposition 4.11 of the source gives

$$V^{\text{hsc}} = (\mathcal{S}_1(H) \cap V^{\perp})^{\perp} = (\mathcal{S}_1(H) \cap (1-e)K)^{\perp}, \tag{4}$$

where both orthogonal complements are taken in K.

Let A' be the commutant of the displayed action of A on K. Trace-class operators form a two-sided operator ideal, hence D is invariant under the left and right bounded multiplications comprising A'. Since $e \in A$, the space $(1-e)K$ is also A'-invariant. Consequently W_e is invariant under the self-adjoint algebra A' and is therefore reducing for A'. Its orthogonal projection r_e commutes with A', so

$$r_e \in (A')' = A.$$

Equation (4) now says that $\operatorname{Ran}(e^{\text{hsc}}) = W_e^{\perp}$, proving (2).

It remains to identify r_e as the join in (1). For any $\xi \in K$, the projection onto $\overline{A'\xi}$ belongs to A and is the smallest projection $p \in A$ satisfying $p\xi = \xi$; this is precisely $s_A(\omega_\xi)$. The set $D \cap (1-e)K$ is A'-invariant. Hence

$$W_e = \overline{\operatorname{span}}\{\overline{A'\xi} : \xi \in D,\ e\xi = 0\},$$

whose projection is the join of the corresponding support projections. Substitution in (2) proves (1). Finally, $e = e^{\text{hsc}}$ is equivalent to $r_e = 1-e$, which is exactly (3). $\square$

3 The full-factor and finite-corank cases

Corollary 2. *Suppose $M = \mathcal{B}(H)$ in its defining representation. Then $A = \mathcal{B}(K)$ and, for every projection e on K,*

$$e^{\text{hsc}} = 1 - P_{\overline{\ker(e)\cap\mathcal{S}_1(H)}^{\|\cdot\|_2}}. \tag{5}$$

If $F = \ker e$ is finite-dimensional, then

$$e^{\text{hsc}} = 1 - P_{F\cap\mathcal{S}_1(H)}. \tag{6}$$

In particular, when $F = \mathbb{C}\xi$,

$$e^{\text{hsc}} = \begin{cases} e, & \xi \in \mathcal{S}_1(H), \\ 1, & \xi \notin \mathcal{S}_1(H). \end{cases}$$

Proof. For $M = \mathcal{B}(H)$, the standard action gives $M \bar{\otimes} M^{\text{op}} = \mathcal{B}(\mathcal{S}_2(H))$; thus Theorem 1 immediately gives (5). If F is finite-dimensional, its linear subspace $F \cap \mathcal{S}_1(H)$ is closed in F, so the closure in (5) may be removed. In dimension one the intersection is either all of F or $\{0\}$. $\square$

The source's Example 4.12 is the second alternative: its normal vector is the diagonal Hilbert–Schmidt operator with diagonal $(1/n)$, which is not trace class, and the closure fills the entire diagonal defect. The corollary shows that this phenomenon is universal for every corank-one projection, not special to a diagonal model.

4 Commutative interpretation and remaining gap

For the standard representation $M = L^\infty(X) \subseteq \mathcal{B}(L^2 X)$, the algebra A identifies with $L^\infty(X \times X)$ on $K = L^2(X \times X)$. If $e = 1_E$, formula (1) says that $1 - e^{\mathrm{hsc}}$ is the indicator of the essential union of supports of all trace-class kernels supported in E^c. Here the trace-class kernels are the image of $L^2(X)\widehat{\otimes}_\pi\overline{L^2(X)}$ in $L^2(X \times X)$. For a countable atomic X, the matrix units show that every projection is fixed. In the diffuse case the fixed-point problem is a form of operator synthesis: which measurable complements are generated, in L^2, by the trace-class kernels they support?

Thus Theorem 1 is an exact projection-lattice description and gives nontrivial complete special cases. What remains open is a representation-free characterization of the join in (1) using only intrinsic structure of an arbitrary $M\bar{\otimes}M^{\mathrm{op}}$, or an equally explicit classification of its fixed projections. The paper's separate abstract Hilbert-module reconstruction question for operator-valued weights is not addressed here.

5 Novelty note

The proof is a short consequence of Proposition 4.11 once support projections are brought into the picture. The source does not state formula (1) or the finite-corank classification. Searches of the local run corpus through 11 August 2026 found no duplicate result, but the argument may be regarded as an implicit reformulation by specialists. Novelty confidence is therefore moderate, while mathematical-validity confidence is high. Independent expert review is requested.

References

[1] M. Daws, *Quantum graphs in infinite-dimensions: Hilbert–Schmidts and Hilbert modules*, arXiv:2511.23121v1 (2025), https://arxiv.org/abs/2511.23121.